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An experimental study to validate optimum distance between metal hydride tanks with staggered arrangement for effective thermal management

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HIGHLIGHTS

- A numerical model for transferring thermal energy efficiently among multiple tanks is validated.
- The usage of optimum spacing among MH tanks for transferring thermal energy efficiently among multiple tanks are studied.
- The 5 and 6 mm distances applied with $Re = 12{,}000$ and 6000 respectively are experimentally proved to be the most ideal.
- The usage of metal hydride tanks with fin with the ideal distance might be critical in future works.

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ABSTRACT

Thermal control is crucial in enhancing the life cycle of metal hydride tanks, which necessitates considering several critical parameters. In addition, a suitable arrangement of a metal hydride bank plays a significant role in transferring the heat efficiently. We have already studied the existence of an optimal distance in theory and proved that the maximized heat transfer rate is a function of the geometric arrangement. In this study, which is the second step of our previous work, we aim to experimentally validate the possible ideal distance in optimized metal hydride array by taking into account the results of our previous study. The experiments are carried out at a triple metal hydride bank with a forced convection system and the effects of optimum distance on the heat transfer performance of the metal hydride bank are analyzed with the staggered arrays in order to validate the results of numerical calculations for staggered metal hydride bank with the pitch-to-diameter ratio of 0.06 and 0.07 at the Reynolds number of 6000 and 12,000, respectively. The results obtained at the end of the experiments are as follows; *i)* metal hydride storage tanks are strongly affected by the array geometry for maximum heat transfer, especially for small pitch arrays at high Reynolds numbers; *ii)* the experimental findings are compared with the theoretical models and validated; *iii)* the overall efficiency of fuel cell-metal hydride system could potentially be increased by simple geometrical arrangements.

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Introduction

The role of renewable energy sources such as wind, solar, geothermal, tidal, biomass, and hydro is crucial in significantly reducing carbon emissions. Hydrogen is also another renewable form of energy being that its high energy content per unit weight. However, this very reactive element does not exist as a single atom in nature. For example, while it exists in the plasma state at stars like the Sun in the universe, most of the hydrogen on Earth is found in molecular forms like water and organic compounds (*i.e. coal as in solid state, natural gas as in gas state, and petroleum as in liquid state*). Being able to be produced from diverse and aforementioned domestic resources makes hydrogen another crucial form of renewable energy source. With these in hand, hydrogen-based renewable energy production is planned to be *19 EJ* in 2050 to achieve the global energy transformation. In addition, the generated hydrogen can be compressed and stored in a storage module for several months [1]. However, the cost of stored hydrogen-based energy is currently too high and it doesn't seem to reduce at the same rate as in renewable power investment costs. Several steps in-hierarchy are required to ensure that effective storage of hydrogen is possible. These steps are profoundly reflected in many studies that are shared by varying researchers [2].

Efficient energy storage is critical for maintaining energy sustainability and reducing the losses due to ill-advised design. Today, owing to the fact that the cost increase in energy demands, stored hydrogen is considered a solution for automobile and transportation related applications thanks to the cost-effectiveness of liquid hydrocarbons. Additionally, energy stored in the bonds of chemical compounds has become one of the most widely used energy storage methods for the last century. As stated above, since hydrogen tends to be in compound form, it is also possible to store it chemically within metal hydride. However, metal hydrides have several disadvantages, that is low-density energy on a mass basis in regard to the weight of the metal powder, in storing hydrogen. On the other hand, metal hydride has high-density energy on a volume basis. Thus, this kind of stored energy provides reliable advantages over the other methods such as compactness and a wide variety of metal hydrides operated at different temperature-pressure according to characteristics of the storage system.

Metal hydrides are extensively used in hydrogen-fueled automobile and transportation applications because of their good energy density and low operating pressure. It is very well known that the performance and life cycle of metal hydrides are highly related to temperature. Additionally, the extra heat transfer area has an impact on heat management, and more than one tank is preferred as a solution. Moreover, when one tank is used, the system's performance will be affected by the performance of the metal hydride tank. Therefore, it is important to utilize more than one tank in the system in case it is accomplishable. In addition, occupied space and investment costs should be considered and competed economically with cost-effective hydrocarbons for mobile and transportation applications – for providing affordable and reliable vehicles to end-users.

A wide variety of experimental studies have been released in the literature for the last years to enhance the heat transfer and thus provide a stable hydrogen capacity. However, no focus on the effect of array geometry of metal hydride tanks has appeared in these publications. All studies on this specific subject have been examined with consideration of a single tank. Omrani et al. designed and positioned horizontally the nine metal hydride tanks in two parallel rows in a well-insulated air box [3]. Mahmoodi et al. found that the desorption performance of a metal hydride tank using the new novel geometric configuration of heat pipes might be experimentally increased. Additionally, they focused especially on the effects of heat pipe number and fin in metal hydride tanks [4]. Kumar et al. studied the absorption/desorption characteristics of a metal hydride tank with 99 cooling tubes [5]. Tarasov et al. used multiple metal hydride tanks in experiments [6]. Lototskyy et al. developed an assembly of the metal hydride containers staggered with heating/cooling tubes. But the study does not contain any data related to tank arrangements [7]. A 3D, kinetic, numerical, and physical model has been developed, simulated, and experimented by Lin et al. in order both to optimize metal hydride tanks and to increase the hydriding/dehydriding rate by applying different methods such as pulverization, self-densification, constant flows for the gas-solid reaction and particle and porosity [8–11]. Hilali et al. studied the theoretical arrangement of metal hydride tanks under different conditions. The study examined an optimum distance among tanks based on relevant conditions [12].

The purpose of our study is to experimentally prove that an ideal distance might be possible in a metal hydride array by considering the results of the theoretical study with two main parameters, which are kinetic equation for the desorption process and heat equations. The tanks are placed in the theoretically calculated distances and experimentally examined in this study.

Theoretical study

Three metal hydride tanks filled with $LaNi_5$ alloy are used, and the staggered arrangement is considered as illustrated in Fig. 1. U_∞ and T_∞ are described as air velocity and air temperature respectively.

The tanks are selected as solid cylinders of equal size by simplification. The Van't Hoff plots for describing the relations between the hydrogen equilibrium pressure and the temperatures according to the metal hydride types are used for numerical modeling as shown in Fig. 2. The Van't Hoff plots indicate the thermodynamics of a hydride, which leads to the reaction enthalpy and entropy. $\ln P_{eq}$ correlates linearly with 1/T; where the slope and Y-intercept correspond respectively to the enthalpy of reaction and to the entropy of the reaction, estimating the straight line to the Y-axis intersection [13–16]. The equilibrium pressure is calculated by Eq. (1):

$$\ln P_{eq} = A - \frac{B}{T} \tag{1}$$

$$T_w = T = \frac{B}{A - \ln P_{eq}}$$

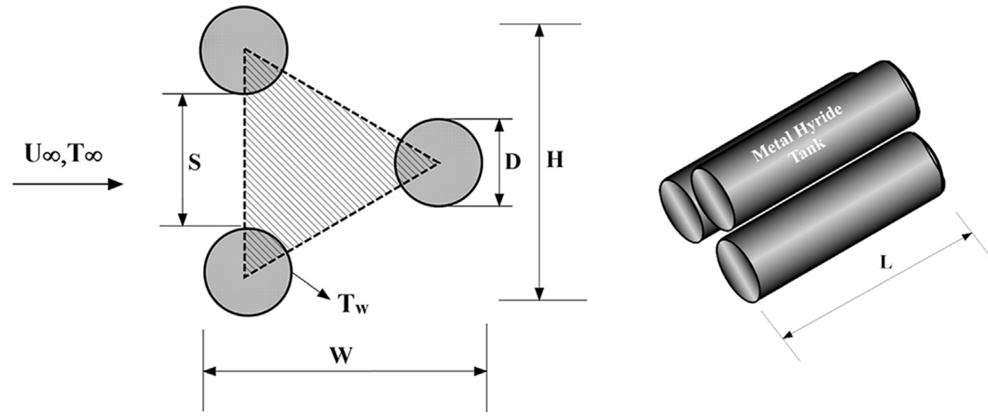

Fig. 1 – Geometric definitions of the staggered arrangement.

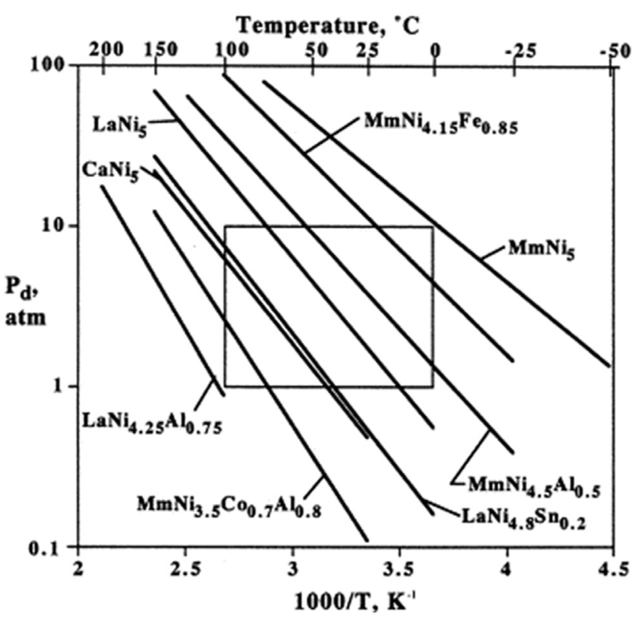

Fig. 2 – The Van't Hoff plots of some metal hydrides.

where constants A and B represent the enthalpy and entropy change in the kinetic origin of $LaNi_5$ alloy.

The intersection of asymptotes is illustrated in Fig. 3 with the purpose of providing the existence of an optimal distance in maximizing the heat transfer rate. This technique was used to optimize the geometric arrangement. The aim was to find the relationship between the heat transfer rate Q, and the varying spacing, S, in the two extreme distances at $S \rightarrow 0$ and S → □. Then the formulas are equalized to obtain the ideal distance [17].

The heat transfer rate, that is q_{large}, is calculated as in the following equations from Eq. (2) to Eq. (6). The equations start with the calculation of the extreme limit of a maximum distance as shown in Eq. (2);

$$q_1 \cong \frac{k}{D} Nu_D \pi D L (T_w - T_\infty) \quad (2)$$

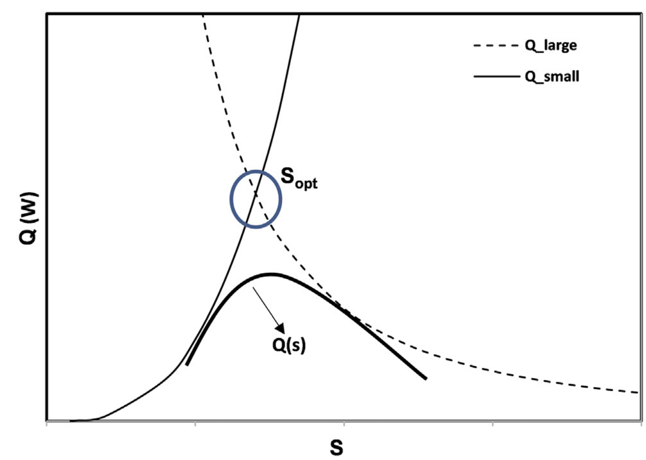

Fig. 3 – The ideal distance as the intersection of asymptotes.

where the Nusselt number is calculated using the following formula:

$$Nu_D = 0,71 \cdot Re_D{}^{0,6} \cdot Pr^{0,36} \; Re_D \geq 10^3 \quad (3)$$

$$Re_D = \frac{U_{max}.D}{\nu}, \quad U_{max} = \frac{(S+D)}{S}.U_\infty$$

In Eq. (3), the maximum velocity occurring in the staggered tank configuration defines the flow characteristics. In this formula, outer diameter D and υ denote the hydraulic diameter and the viscosity of the fluid respectively at the film temperature ($T_f = (T_w - T_\infty)/2$).

The number of Hydrogen tanks in the cross-flow plane is found with the following formula:

$$n = \frac{H\,W}{(S+D)^2 \mathrm{Cos}\;30} \quad (4)$$

The heat transfer rate is calculated by

$$q_{large} = q_1 * n \quad (5)$$

$$q_{large} \cong 2,58 \frac{HLW}{(S+D)^2} \, k(T_w - T_\infty)\{Re_D{}^{0,6} \cdot Pr^{0,36}\} \quad (6)$$

From Eq. (6) for a maximum distance, the maximum heat transfer rate is inverse proportional to S^2 and decreases while the distance is becoming larger.

In the opposite extreme limit of minimum distance, the heat transfer rate, q_{small} is defined in terms of Reynolds number as:

$$q_{small} = \dot{m}\, c_p (T_w - T_\infty) \quad (7)$$

where $\dot{m}$ is the total mass flow rate as calculated through the L × W plane, $\dot{m} = \rho WLU$. U is defined as a mean velocity and derived by using Darcy and Carman-Kozeny equations [18].

$$U \cong \frac{\rho U_\infty^2 D^2}{200\mu H} \frac{\left[\left(S/D\right)^2 + 2S/D\right]^3}{\left[1 + S/D\right]^2} \quad (8)$$

$$q_{small} \cong \frac{1}{25} \rho . C_p . \nu . \frac{W.L}{H} . Re_D^2 . \left(\frac{S}{D}\right)^3 . (T_w - T_\infty) \quad (9)$$

From Eq. (9) for a minimum distance, the maximum heat transfer rate is proportional to S^3 and increases while the distance is increasing.

The optimum arrangement could be achieved by combining Eq. (6) and Eq. (9) using the intersection of asymptotes as shown in Fig. 3.

$$q_{large} = q_{small}$$

The optimization result is:

$$\left(\frac{S_{opt}}{D}\right) \cong 4.Re_D^{-0.47}.Pr^{-0.21} \quad (10)$$

The ideal distance is predicted by the correlation given in Eq. (10). The maximum heat transfer rate can be obtained for a defined volume that corresponds to the ideal distance, S_{opt} by substituting Eq. (6) or Eq. (9).

The numerical analysis of our system was simulated by using the finite element method in a CFD Program (i.e., Autodesk CFD) due to its flexibility in modeling any geometric shape. The first validation was compared with the outputs of the analytical results.

Experimental work and validation

In order to validate the proposed model, an experimental setup is designed as illustrated in Figs. 4 and 5. Moreover, measurements were calculated based on the same parameters, namely air velocity, air temperature, and geometric configurations. The setup consists of seven components namely: an axial fan with frequency controller (1), an air heater with accurate and sensitive PID temperature controller (2), an airflow meter (3), a fuel cell system (4), metal hydride storage tanks with an outer diameter of 85 mm and a length of 400 mm (5), black coated aluminum plates (6) and a thermal imaging camera (7). Thermal images are recorded for temperature distribution between tanks by the thermal camera (i.e. Testo 885 professional thermal imager), which is used precisely to visualize and to the analysis of captured images in the experiments [19]. It is also important to note that the dimension of the duct is specified as 40 cm × 40 cm × 300 cm (W × H × L). The aluminum plates are perforated to the diameter of the metal hydride tanks and are mounted separately in the channel during the experiments. In the case of a perforated plate, the perforations form a gap between the tanks. The professional thermal imaging camera is mounted at a distance of 1 m from the metal hydride storage tanks during the experiments. The fuel cell system is based on the 1200 W Nexa power source including an energy management module and electronic charger, as shown in Fig. 2 [20].

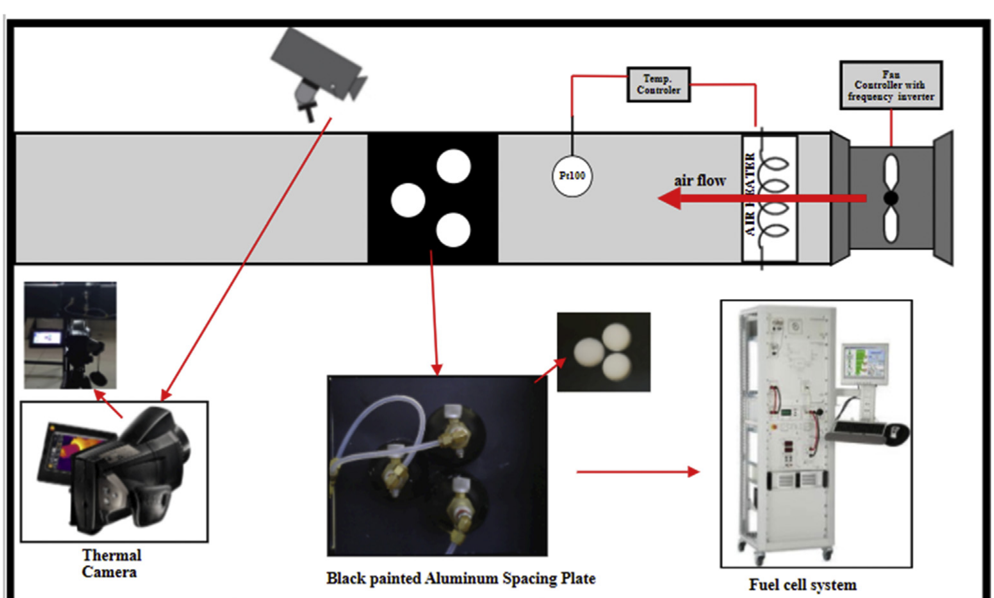

Fig. 4 – Schematic of the experimental setup.

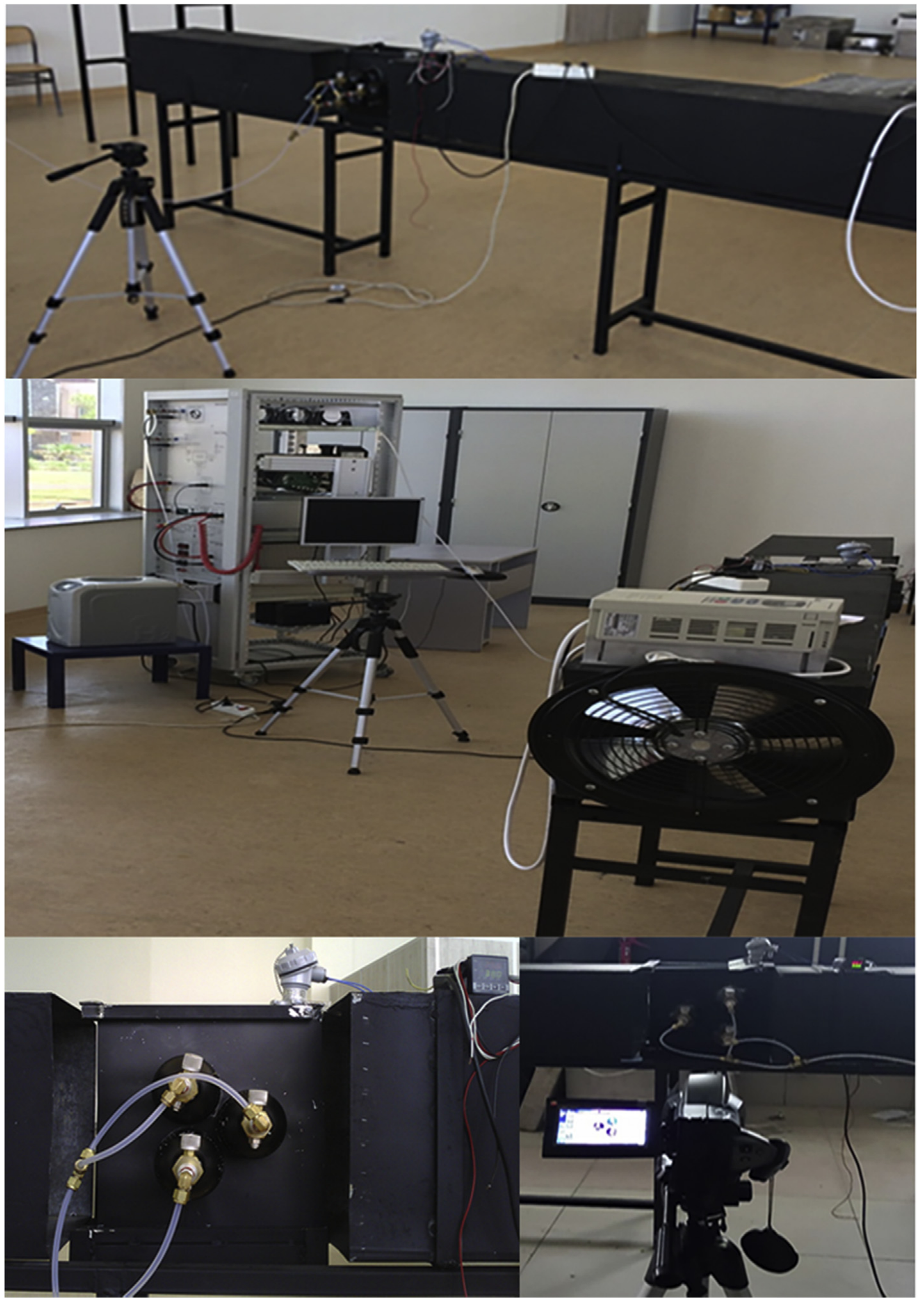

Fig. 5 – The principal elements of the test setup.

The thermal imaging camera is calibrated according to factory specifications. Also, before the experiments, the accuracy of the camera is verified by measuring objects with known temperatures.

The main findings of the study might be emphasized in two parts: The first step is to demonstrate the possibility of integrating the proposed arrangement with the fuel cell, and the second step is to observe the working regime of the system. The hydride tanks are fully charged by the electrolyser before the experiment is started. The designed system is operated for about 100 min during the experiment until the desired equilibrium pressures are reached by using the fuel cell. During the experiments, tank pressures are measured by a fuel cell energy management module. The present study consists of two stages determined by perforated plates; each of them evaluates the heat transfer rate separately based on different ambient temperatures (300 K & 310 K), Reynolds numbers (6000 & 12,000), and equilibrium pressures (60 kPa & 120 kPa). In each of the two stages, two plates are used with a distance of 5 and 6 mm between the tanks. These plates are selected based on the findings obtained in the previous results of the theoretical study. The tank pressures are obtained by setting the fuel cell up to the specified values under the constant load using an electronic loader. In the first stage, the plate is mounted to the duct with a distance of 5 mm between the tanks, and metal

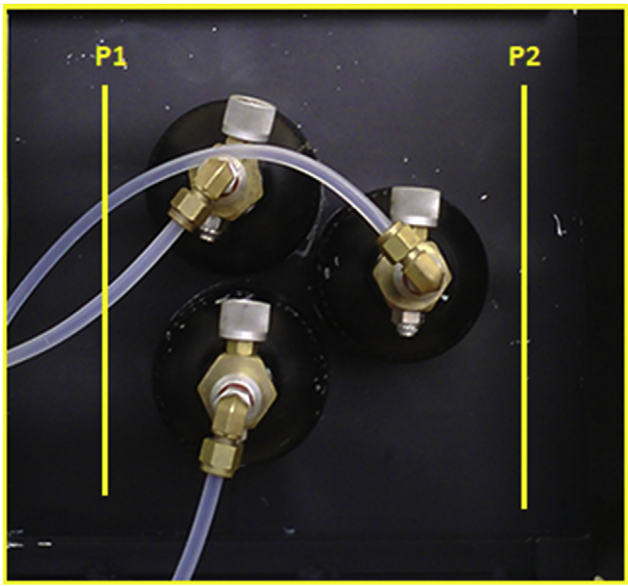

Fig. 6 – Measurement planes on the face of the plates.

hydride tanks are placed in the plate. Then, as mentioned above, different ambient temperatures, Reynolds numbers, and equilibrium pressures are implemented in the setup for optimal heat transfer. The temperature on the plate is recorded with a thermal imaging camera while the metal hydride tanks are discharged under load during fuel cell operation. In the second stage, the same procedures are repeated for the plate with a distance of 6 mm.

Results and discussion

As shown in Fig. 6, the air temperature difference (ΔT), the difference between the inlet (P1) and outlet (P2) air temperatures, is defined in Eq. (11).

$$\Delta T = T_{p1} - T_{p2} \tag{11}$$

The heat transfer rate is expressed as

$$Q = \dot{m}_{air}\, C_p\, (T_{p1} - T_{p2}) \tag{12}$$

Here, $\dot{m}_{air}$ and C_p denote the airflow rate and the specific heat capacity of air, respectively.

As stated earlier, the temperature distribution on the plate surfaces and tanks are visualized and recorded by utilizing the thermal imaging camera. The captured thermal images are used to determine the effects of optimal distance on maximum heat transfer. The solid lines drawn on the plate in Fig. 6 are used to plot the temperature profile using the thermal image data along the lines to compare the results of the theoretical study. The contours in Fig. 7 illustrate the temperature distribution over the plate area based on the solid lines for relevant variables. The results of CFD and the experimental results are shown on the left side and the right side of Fig. 8, respectively.

Fig. 7 illustrates the temperature distributions on the surface of the plate along each vertical line (P1 & P2) at $P_{eq} = 60$ kPa and $P_{eq} = 120$ kPa, respectively. The blue and red counter indicates the inlet and outlet air temperature respectively. The difference between the blue and red counters indicates the heat transfer rate of the metal hydride tanks. As the figures show, the temperature difference is approximately between 0.5 and 1 K depending on the Reynolds number. When the Reynolds number increases, the temperature difference also increases. These results are similar to the findings of another study of ours.

Fig. 8 illustrates the results obtained by both simulation and experiments. As illustrated, both results are highly similar and verify the validity of the model. As seen, both contours are similar and the small differences are a consequence of the thermal camera measurements and the many uncertainties associated with the experimental setup since the simulated values are calculated under ideal conditions. In addition, different tank temperatures in

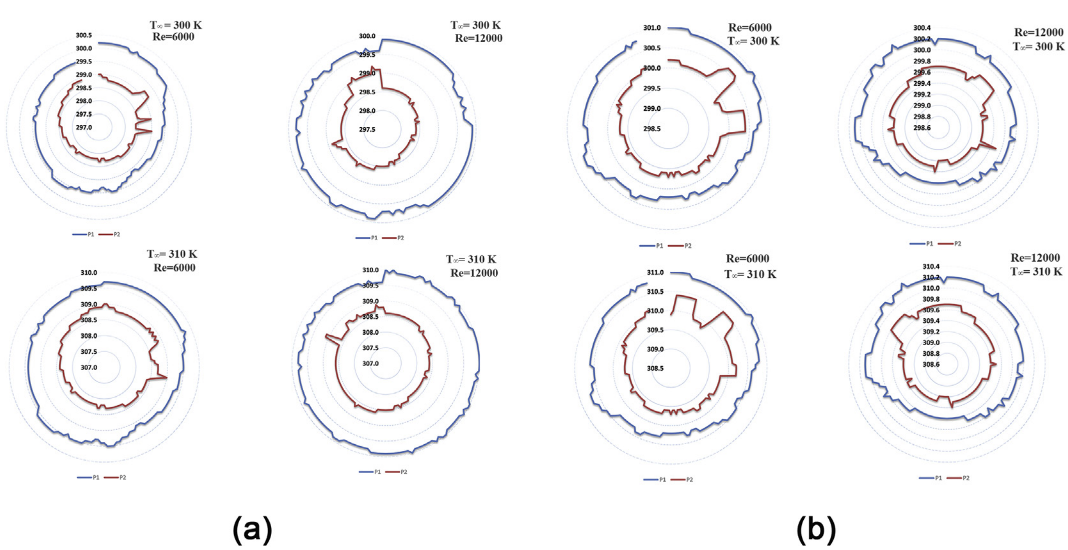

Fig. 7 – The temperature profiles captured on the face of plates for (a) $P_{eq} = 60$; (b) $P_{eq} = 120$ kPa.

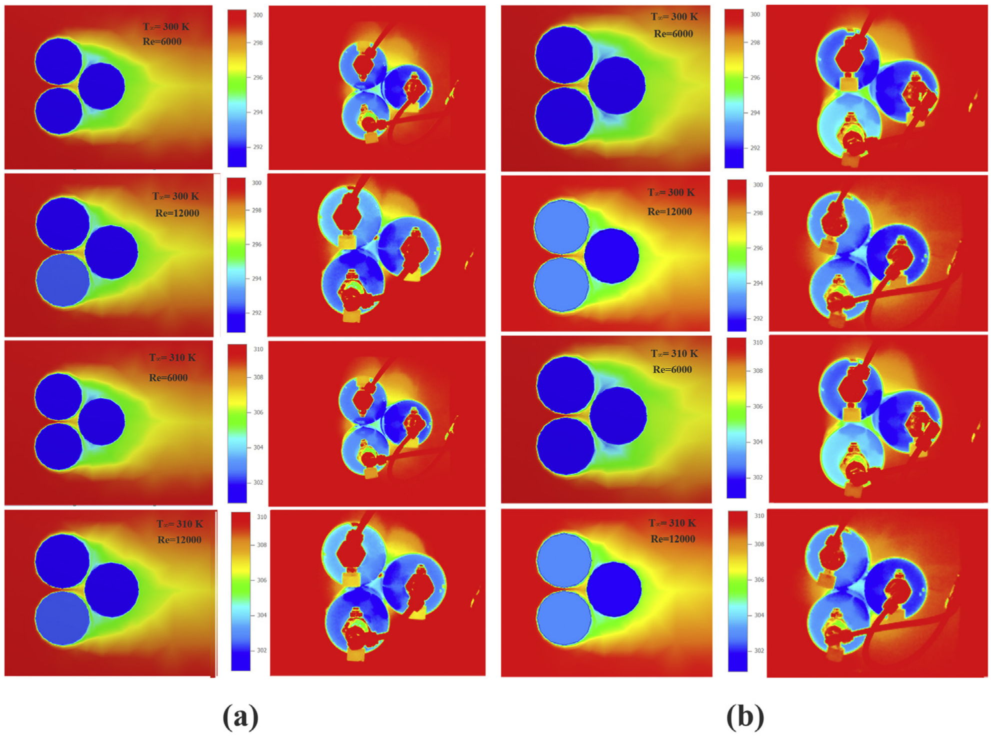

Fig. 8 – The visual comparison of CFD results and experimental results (a) for P_{eq} = 60 kPa; (b) for P_{eq} = 120 kPa.

Table 1 – The comparison of numerical and experimental heat transfer rate.

P_{eq} (kPa)	T_w (K)	Re = 6000			Re = 12,000		
		S_{opt} (m)	$Q_{numerical}$ (W)	$Q_{experimental}$ (W)	S_{opt} (m)	$Q_{numerical}$ (W)	$Q_{experimental}$ (W)
T_∞=300 K							
60	274.1	0.006	−349	−311	0.005	−539	−498
120	289.0	0.006	−147	−120	0.005	−227	−199
T_∞=310 K							
60	274.1	0.006	−481	−311	0.005	−742	−498
120	289.0	0.006	−279	−203	0.005	−430	−349

experiments are also related to the kinetics of the hydriding-dehydriding reactions.

Table 1 proves that the theoretical heat transfer rate is validated by the experimental results. As seen in Table 1, the experimental heat transfer rate varies from ~120 to ~500 W for defined pressures and temperatures. Similarly, the theoretical heat transfer rate varies from ~150 to ~750 W.

The pressure variation curves of the metal hydride tank array according to time at different Reynolds numbers, and equilibrium pressures are illustrated in Fig. 9. The optimum distance curves found in the theoretical study are attached to this figure. Two distances (i.e. 5 and 6 mm) that one of which is the optimum distance, are chosen for pressure comparison according to the attached figures. As illustrated, the pressure drop rate of the optimum distance is the slowest and that of the non-ideal distance is the fastest at four experiments. The results show that the pressure performance of the ideally arranged tanks is the best while the non-ideally arranged tanks are the worst. In addition, it should be noted that when the tanks are at 60 kPa of equilibrium pressure, ideal and non-ideal distance curves indicate a wide time difference (9–13 min), while ideal and non-ideal curves show a less time difference (2–3 min) at 120 kPa of equilibrium pressure. Fig. 10 shows calculated fractions of the experimental heat transfer rate to the theoretical heat transfer rate as a function of Reynolds number. The calculations are performed for a temperature range from 300 to 310 K. Comparing the experimental results with the theoretical ones, it can be seen that the best result is obtained for P_{eq} = 60 kPa and *Re* = 12,000 at 300 K and the worst for P_{eq} = 60 kPa and *Re* = 6000 at 310 K. On the other hand, the heat transfer fraction indicates that while the rate is increasing, experimental results (e.g. 92%) are becoming more meaningful.

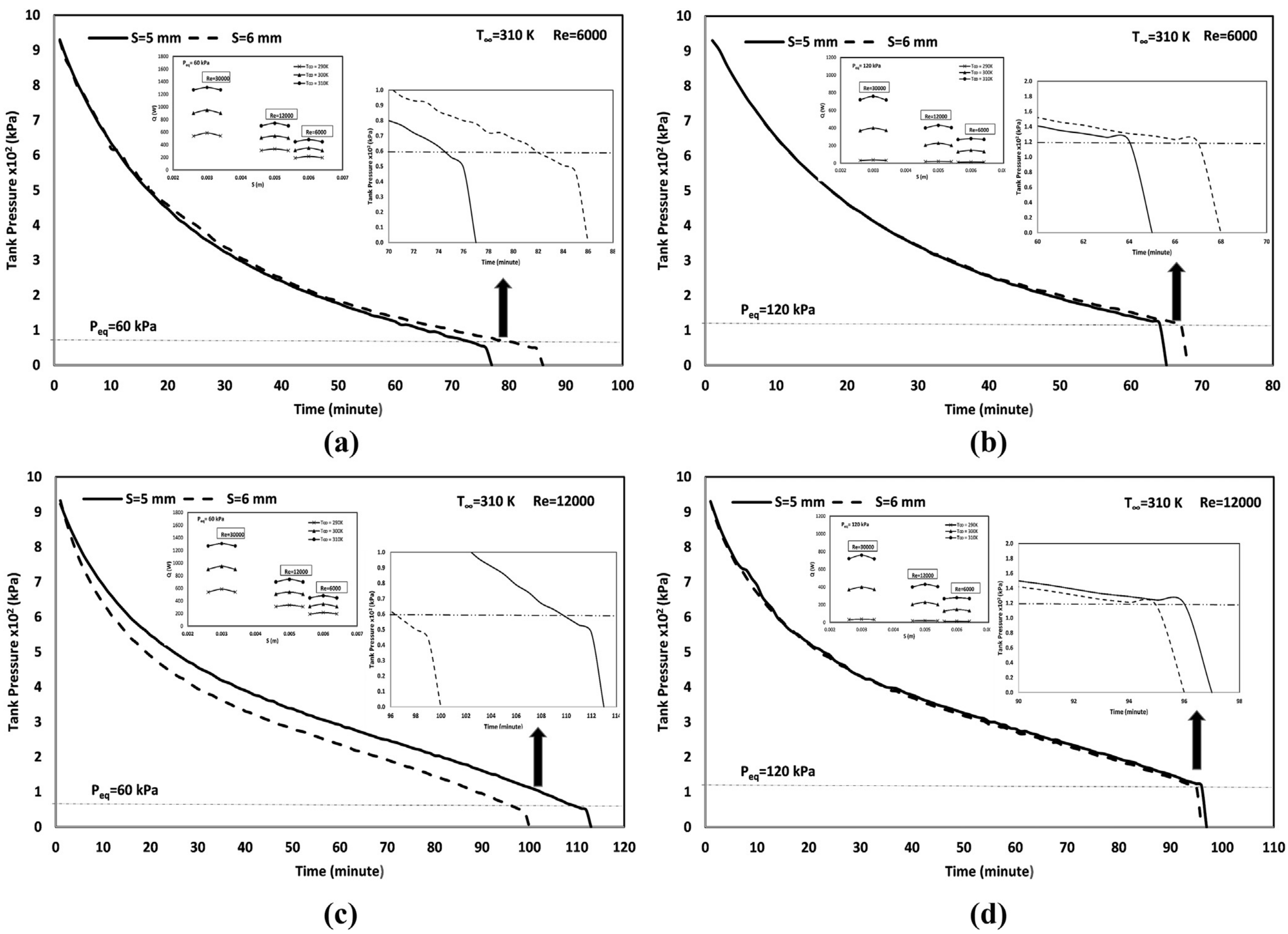

Fig. 9 – Diagrams of tank pressure change (a) and (b) for $Re = 6000$, (c) and (d) for $Re = 12,000$.

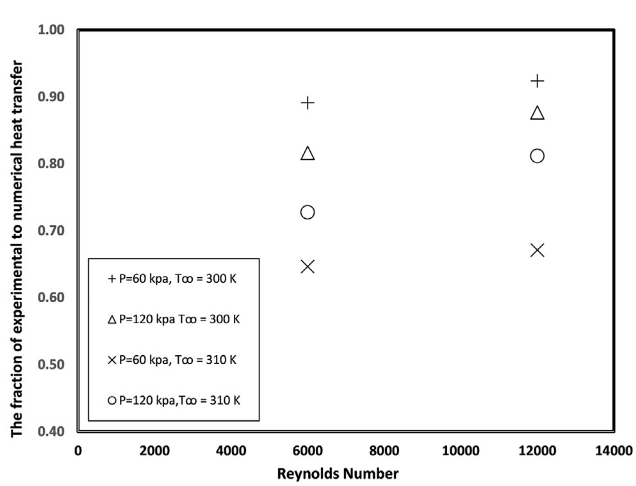

Fig. 10 – The fractions of experimental heat transfer rate to the numerical heat transfer.

Conclusions

This work focused on the cruciality of optimum distance among metal hydride tanks for transferring thermal energy efficiently. A numerical model is used to develop an experimental study to validate optimum distance among metal hydride tanks with staggered arrangements for effective thermal management. For experimental validation of the recommended mathematical model, heat transfer during discharge and temperature distribution were considered. The varying inputs such as air velocity, pressure, and ambient temperature were applied as main parameters to affect the temperature distribution during discharge. The data output as the result of the thermal response of input parameters revealed the importance of optimizing an ideal distance between the metal hydride tanks, as it plays a critical role in the ability to transfer heat above 75% of the theoretical capacity. Therefore, the study concludes that an optimal distance between the metal hydride storage devices has positive effects on the thermal conductivity of the proposed system as it critically improves effective thermal management. Moreover, the usage capacity of the tanks is to increase related to ideal distance. The heat transfer losses may occur outside the measurement range due to the design of the setup. In future studies, it can be fixed, and a more effective result can be achieved by using more than three tanks and different metal hydride materials. The experimental and theoretical results of this work can provide a direction for the characterization of a metal hydride storage system. Therefore, an effective design of the storage system can be implemented. The placement alternatives can be suggested according to the number of metal hydride tanks and material (type of metal hydride alloy and tank material), or an indication can be given on how to improve hydrogen discharge depending on constrained conditions. In addition, it is important to consider the potential of contributing to commercially available works such as automobiles, etc. In the mobile systems powered by fuel cells, the distance could be arranged by software and mechanical systems according to the vehicle speed and could validate the concept of autonomous thermal energy management. This study shows a promising future for technology and research in this field.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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